duet-tools: A Python package for interacting with the DUET program

Niko J. Tutland[a,*]

[a] *New Mexico Consortium, Center for Applied Fire and Ecosystem Science, Los Alamos, NM, USA*

Abstract

Fuel modeling is a key avenue for understanding dynamics of fire-vegetation interactions across landscapes. Three-dimensional (3D) fuel models are used as inputs for fire behavior models to develop strategies for prescribed fire application and wildfire risk assessment and mitigation. Distribution of Understory using Elliptical Transport (DUET) is a recently developed program for creating surface fuel inputs for 3D fuel models. DUET simulates litter fall from 3D tree canopy inputs and grass growth, producing spatially heterogeneous estimates of fine fuel distribution and characteristics in forested domains. Here we introduce `duet-tools`, a Python package that streamlines the process for creating DUET simulation input files and calibrating the values in DUET output files. The package handles two primary aspects of the DUET workflow: (1) programmatic creation and management of the DUET input file with validation and documentation, and (2) calibration of fine fuel outputs towards values provided by online data sources or directly from the user. By simplifying these tasks, `duet-tools` allows modelers and managers to more easily interact with the DUET program while incorporating locally accurate fuels data.

Keywords: Python, fuel modeling, understory, fire modeling, simulation, fire behavior

Metadata

1. Motivation and significance

In landscapes across North America, wildland fire is a critical driver of ecosystem processes that can both maintain and disturb habitats [1]. Modeling

*Corresponding author

Email address: `ntutland@gmail.com` (Niko J. Tutland)

Nr.	Code metadata description	Metadata
C1	Current code version	v1.0.1
C2	Permanent link to code/repository used for this code version	`https://github.com/nmc-cafes/duet-tools`
C3	Permanent link to Reproducible Capsule	—
C4	Legal Code License	MIT License
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python (NumPy, pandas, rasterio, landfire)
C7	Compilation requirements, operating environments & dependencies	Microsft Windows, Linux; Python $\geq$ 3.10; dependencies installed via pip
C8	Link to developer documentation/-manual	Full documentation, how-to guides, example scripts, and issue tracking: `https://nmc-cafes.github.io/duet-tools/`
C9	Support email for questions	ntutland@gmail.com

Table 1: Code metadata

wildland fire behavior is a key tool for planning land management practices like prescribed burning, where complexities in fire-atmospheric interactions are mediated by heterogeneous, three-dimensional (3D) fuel structures. Physics-based fire behavior models, such as FIRETEC [2] or QUIC-Fire [3], simulate fire behavior in 3D, but take in relatively fine-scale (<4 m^3) inputs describing fine fuel density, size, moisture, and height. In forested landscapes, modeling the distribution of fine fuels on the ground surface (e.g., leaf litter, herbaceous fuels) is challenging at these scales because of their heterogeneity and the difficulty of observing them with survey tools such as airborne LiDAR.

The Distribution of Understory using Elliptical Transport (DUET) program [4] presents a solution by simulating the spatial arrangement of leaf or needle drop based on tree crown dimensions and prevailing wind direction, placing grass where litter levels would not limit its growth. However, these simulations may not capture some essential elements of understory dynamics that vary geographically, such as decay rates, herbaceous fuel loading, or fuel moisture content. Thus, users of DUET may wish to modify outputs of the simulations, differentially shifting the magnitudes of loading, height, and/or moisture values based on field-measured values, expert opinions, or broad-scale datasets, without altering the spatial distributions predicted by DUET.

Despite the potential for DUET to be used to improve 3D fuel modeling in forested landscapes, no standardized tool yet exists for managing the inputs and outputs of DUET simulations to help parameterize realistic, place-based 3D fire behavior simulations.

Here, we introduce the `duet-tools` Python package, which addresses these needs by providing: (1) a simple, validated interface for creating, reading, and modifying DUET input files; (2) intuitive management of DUET output arrays; (3) built-in functions for calibration of fuel elements; and (4) comprehensive documentation with how-to guides and example scripts.

2. Software description

2.1. Software architecture

`duet-tools` is implemented in Python using a modular design with a combination of functional and object-oriented features. Functions are applied to two main classes representing the package's functionalities: (1) the `InputFile` class, which manages DUET input files by loading, creating, and writing them with validation; and (2) the `DuetRun` class, which organizes DUET outputs, enabling calibration, format conversion, and export to FIRETEC or QUIC-Fire formats. Typically, a DUET simulation will be run locally in a directory containing the DUET executable file and a text file specifying simulation parameters called `duet.in`. In `duet-tools`, the `InputFile` class represents the parameters within `duet.in` for easy programmatic interaction, and additionally provides the information for automatically importing the data outputs of a completed DUET simulation. These data outputs, which are stored in idiosyncratic data structures, can then be read and represented by a `DuetRun` object.

Once the DUET simulation outputs are imported, `duet-tools` offers flexible methods for modifying or "calibrating" the outputs. Target values and calibration methods for the magnitudes of DUET data can be defined, assigned to various fuel types, and applied differentially based on the specific needs of the user. By following the general workflow of assigning targets, setting those targets for fuel parameters, then calibrating a `DuetRun` object, a user can incorporate local data or expert knowledge into their DUET simulations. The resulting data arrays are represented in standard NumPy [5] formats, which can then be processed manually by the user or conveniently exported to data formats accepted by QUIC-Fire or FIRETEC.

2.2. Software functionalities

The package consists of several modules, three of which we highlight here: (1) `inputs`, (2) `calibration`, and (3) `landfire`. The Inputs module facilitates loading, modifying, and writing validated DUET input files. Input parameters for the DUET program include the 3D fuel array dimensions and resolution, the number of years to simulate, and information on prevailing winds: speed, direction, and variability. The `from_directory()` classmethod automatically creates a representation of `duet.in` which can be conveniently modified and written to a file programmatically. Every parameter is validated upon write, meaning that any errors will be caught before the DUET simulation takes place. This involves type checking and ensuring inputs fall within accepted data ranges.

The Calibration module comprises the bulk of the current functionalities of `duet-tools`, providing methods for shifting and/or scaling DUET output values based on target ranges or distributions. Field data on fuel properties such as fuel loading may be collected and summarized as a range of values or as a distribution with a center and spread. The Calibration module can perform shifting and scaling based on either type of target, and can differentially apply those methods for different fuel parameters and/or fuel types. Targets are assigned using the `assign_targets()` function, which creates an object of class `Targets`. To assign targets based on a range of values, the "maxmin" method applies the following algorithm:

$$x_i' = \begin{cases} \text{min\_target} + \dfrac{x_i - \min(x^+)}{\max(x^+) - \min(x^+)} \cdot (\text{max\_target} - \text{min\_target}), & x_i > 0 \\ x_i, & x_i \leq 0 \end{cases}$$

where $x^+ = \{x_i \in x \mid x_i > 0\}$.

This shifts, then stretches or squeezes the values in the DUET output data array to match the provided range, without modifying the relative relationships between the data. Similarly, calibration based on mean and standard deviation is applied using the "meansd" argument and the following algorithm:

$$x_i' = \begin{cases} \max\left(0,\ \text{mean\_target} + \dfrac{(x_i - \mu_x) \cdot \text{sd\_target}}{\sigma_x}\right), & x_i > 0 \\ x_i, & x_i \leq 0 \end{cases}$$

where $x^+ = \{x_i \in x \mid x_i > 0\}, \quad \mu_x = \text{mean}(x^+), \quad \sigma_x = \text{std}(x^+)$

The user may create one or multiple `Targets` objects which can then be set to different fuel parameters and fuel types. Briefly, a fuel parameter is one of the three required fuel inputs for the QUIC-Fire model: (1) fuel loading, (2) fuel moisture, or (3) fuel depth. A fuel type, on the other hand, corresponds a category of fuel simulated by DUET that each parameter may describe. DUET simulates litter fall from any species of tree represented in the DUET fuel grid input as well as growth of grass or other herbaceous plants. Upon creation of a `DuetRun` object, `duet-tools` groups together the litter from coniferous species, then the litter from deciduous species. This creates three main fuel types to which a `Targets` object may be set: (1) grass, (2) conifer litter, and (3) deciduous litter. Additionally, a target may be set for a combination of both litter types, or to all fuel types together.

The targets corresponding to a single fuel parameter can be set for one or more fuel types using `set_fuel_parameter()`, which creates an object of class `FuelParameter`. Once each desired fuel parameter has been set using targets for each desired fuel type, the `DuetRun` object can be calibrated using the `calibrate()` function. Crucially, any combination of fuel type and fuel parameter can be assigned targets using either calibration method. In practice, this means there is a single function, `calibrate()`, that handles every possible calibration specification. The flexibility to mix and match methods, fuel types, and fuel parameters is facilitated by the use of a small set of keyword arguments in the `assign_targets()` and `set_fuel_parameter()` functions. Running `calibrate()` on a `DuetRun` object produces a new `DuetRun` instance that stores the calibrated arrays. These arrays can be accessed or saved via the classmethods `to_numpy()` and `to_quicfire()`, the latter of which saves the calibrated arrays to a specified directory in the format expected by the QUIC-Fire model.

The `landfire` module supports calibration by offering a convenient source for calibration targets. The main functionality of the module is the execution of a data query of from the LANDFIRE database [6] to be used to calibrate DUET outputs. The module leverages the *landfire* python package [7] to download fuels data, then converts those data to calibration targets. Fuels data themselves are derived from the Scott and Burgan 40 Fire Behavior Fuel Models [8]. Future package development is poised incorporate additional data sources from LANDFIRE, such as the Fuel Characteristics and Classification System (FCCS) [9].

3. Illustrative examples

The main functionalities of `duet-tools` package are compatible with Python version 3.10+ and can be installed using PyPI. Since the `landfire` module requires dependencies that are only compatible with Python version 3.10, it is included as an optional package extra. Full documentation, how-to guides, example scripts, and issue tracking are hosted on GitHub.

3.1. DUET calibration workflow

The use of `duet-tools` is contingent upon the user possessing the results of a DUET simulation. Most often they will exist alongside a DUET input file. A common workflow will begin by creating a `DuetRun` object using `import_duet()`, which derives DUET data dimensions from the `duet.in` input file. If the input file is not present, but the data dimensions are known, the user can alternatively specify `import_duet_manual()`. Next, targets may be assigned using either of the two available calibration methods, and subsequently set to the desired fuel parameter(s). Once targets for each desired fuel parameter have been specified, the `calibrate()` function applies the calibration methods accordingly. A simple example of this process is shown in Listing 1 below.

```
# Import DUET outputs
duet_run = import_duet(directory=duet_path)

# Assign targets for each fuel type and fuel parameter
coniferous_loading = assign_targets(method="maxmin", max=5.0, min=0)
deciduous_loading = assign_targets(method="meansd", mean=0.5, sd=0.1)
grass_loading = assign_targets(method="meansd", mean=0.5, sd=0.25)

# Bring together fuel types for each parameter
loading_targets = set_fuel_parameter(
    parameter="loading",
    grass=grass_loading,
    deciduous=deciduous_loading,
    coniferous=coniferous_loading,
)

# Calibrate the DUET run
```

```
calibrated_duet = calibrate(duet_run=duet_run,
    fuel_parameter_targets=loading_targets)

# Export to a QUIC-Fire input file
calibrated_duet.to_quicfire(quicfire_path)
```

Listing 1: Example of a typical workflow using `duet-tools`, including importing a DUET run, setting calibration targets, calibrating fuel parameter values, and exporting to QUIC-Fire fuel arrays.

In Listing 1, the fuel loading parameter is calibrated using separately defined targets for each fuel type. Grass and deciduous litter loading are calibrated using their respective target means and standard deviations, whereas coniferous loading is calibrated based on a target value range. Figure 1 shows how each calibration shifts the magnitudes of fuel loading values for each fuel type, while retaining the relative relationships between values. In this example, coniferous litter and grass fuel loading have been calibrated to reflect lower loading than was predicted by DUET, whereas deciduous litter loading was increased. After these calibrations, overall loading was decreased, but with patterns corresponding to the modifications of the individual fuel types. The total litter loading is available to be exported as an input file to QUIC-Fire, or as a NumPy array for other applications.

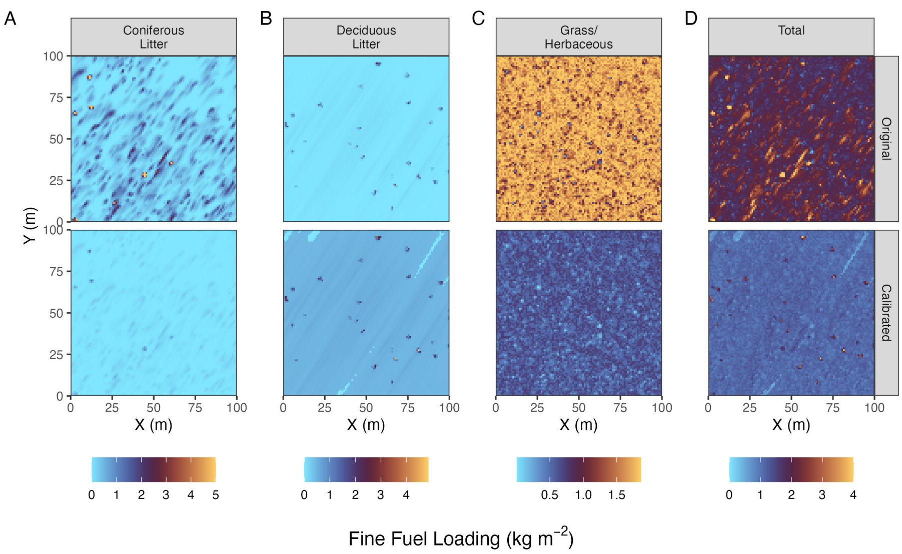

Figure 1: Example of DUET calibration. The top row shows original fuel loading outputs from DUET for each fuel type; the bottom row shows calibrated results based on user-specified targets.

Fuel parameter targets may be assigned from LANDFIRE data in a similar fashion. Listing 2 shows the process of querying the LANDFIRE database to retrieve Scott and Burgan 40 Fuel Model data [8] for an area of interest. Spatial data defining an area of interest can be supplied as either a geojson or a polygon. For each fuel type (grass and/or litter; Fig. 2B), the values of the desired fuel parameters (Fig. 2C) are automatically assigned to a `Targets` object, which can then be used to differentially adjust DUET output values for those values (Fig. 2D-E).

```
# Import DUET outputs
duet_run = import_duet(directory=duet_path)

# Query the LANDFIRE database
landfire_query = query_landfire(
    area_of_interest=aoi_geojson, year=2019, directory=duet_path, input_epsg=4326
)

# Assign fuel targets from LANDFIRE query
grass_loading_targets = assign_targets_from_sb40(landfire_query, "grass", "loading")
litter_loading_targets = assign_targets_from_sb40(landfire_query, "litter", "loading")
landfire_loading = set_loading(
    grass=grass_loading_targets, litter=litter_loading_targets
)

# Calibrate the DUET run
calibrated_duet_landfire = calibrate(duet_run, landfire_loading)
```

Listing 2: Example of DUET calibration using data from LANDFIRE [6]

4. Impact

Accurate descriptions of fuel and vegetation are essential for wildland fire modeling [10], which is an integral aspect of prescribed fire planning and training [11, 12] that supports land management [13], disaster mitigation [14], and species conservation [15]. Simulating realistic fire behavior using tools like QUIC-Fire and FIRETEC requires an understanding of the heterogeneity of fuel structures on a landscape [16, 17], but that heterogene-

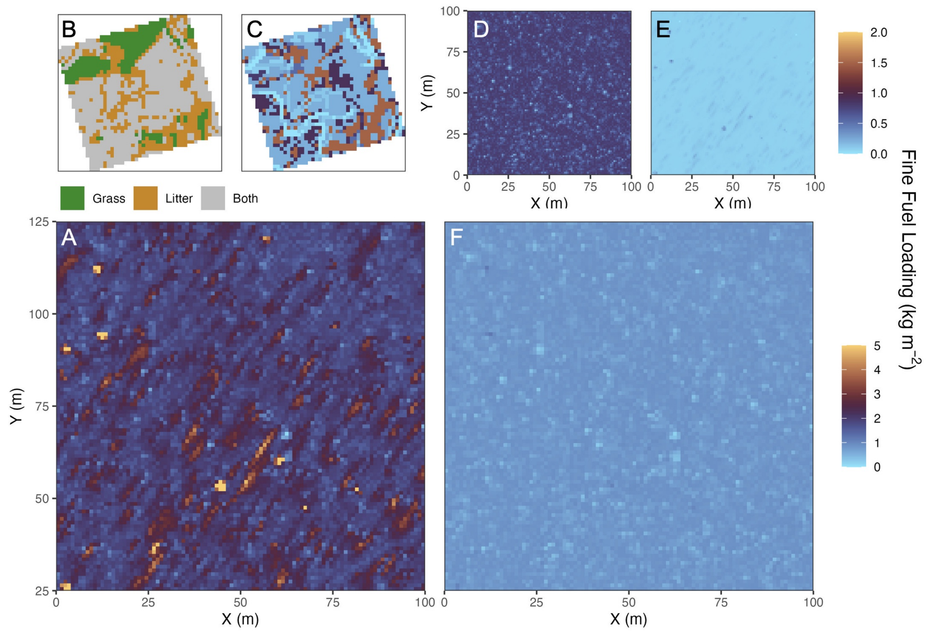

Figure 2: Example of DUET calibration using Scott and Burgan 40 Fuel Model [8] data from the LANDFIRE database [6]. Original DUET output values (A) are assigned fuel parameter targets from a provided area of interest based on the mapped fuel type (B-C). Values for grass (D) and tree litter (E) may be calibrated separately and combined to produce a calibrated DUET output (F).

ity is difficult to characterize. Data fusion approaches that combine local data, broad-scale mapping, and ecological modeling can provide more realistic characterizations wildland fuel heterogeneity [18, 19]. By facilitating the fusion of modeled ecological processes from DUET, satellite-based gridded data from LANDFIRE, and direct input from users, `duet-tools` offers one of the first such approaches for surface fuel modeling.

Though there is utility in fine-scale, data-driven fire behavior modeling, user access to the models, their computational weight, and accurate fuel parameterization stand as barriers to operational use [20]. `duet-tools` helps ease the burden of fuel parameterization by providing a simple and convenient tool for harnessing a fuel modeling software that may otherwise be underutilized. By enabling the programmatic management and modification of DUET inputs and outputs, `duet-tools` has already allowed it to be incorporated into the most robust 3D fuel modeling platform to date, FastFuels [18]. As a national-scale data platform, FastFuels' coverage includes a wide variety of ecosystems, fuel types, and climate zones which can have vastly different litter decay rates [21] and herbaceous growth patterns [22], or may experience a variety of management actions that may modify surface fuels. `duet-tools` provides a key mechanism for accounting for these differences by scaling DUET's predictions to local data inputs, without compromising the spatial heterogeneity predictions that are central to DUET's utility.

As the science of wildland fire management continues to expand its toolkit of computing technologies, tools and frameworks for harmonizing fuel data across scales are becoming more widespread [23, 24, 19]. The continued development of `duet-tools` can help it join the ranks as a way to facilitate this goal, especially as fuel structures and fire dynamics become more uncertain in a changing climate [25, 26, 27].

4.1. Limitations and Future Work

This article aims to introduce `duet-tools`, which is still in the early stages of its development. Currently, manipulation of DUET output files is limited to two simple calibration algorithms and 2-3 fuel types. Future work should incorporate more complex and place-based calibration parameters, such as seasonality [28], topography [29], and the addition of other types of surface fuel such as shrubs. Because local fuel data may not be available, further package development should focus on forging more connections to large geospatial datasets, including the expansion of the `landfire` module to access additional fuel layers, such as FCCS data [9]. Since these layers are generally compiled at a resolution of 20-30 m, the variability they describe is currently applied wholesale when calibrating the fine-scale variability simulated by DUET. Future package development could add the option to incorporate the broad-scale fuel heterogeneity described by broad-scale data while maintaining the fine-scale patterns predicted by DUET simulations. These improvements will help `duet-tools` play an integral role in streamlining the integration of DUET with fuels data platforms such as FastFuels [18].

5. Conclusions

`duet-tools` bridges a key usability gap between DUET and downstream fire behavior models. It simplifies DUET file management, enables automated calibration against data sources, and promotes transparent and reproducible workflows for surface fuel modeling.

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